

CS105 (DIC on Discrete Structures)

Problem set 6

- Attempt *all* questions.
 - Apart from things proved in lecture, you cannot assume anything as “obvious”. Either quote previously proved results or provide clear justification for each statement.
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Part 1

1. Consider the set of all bit-strings, i.e., finite sequences over $\{0,1\}$. EATVA.g., 0101,000,1111110 are all bit-strings.

For any two bit-strings u, v , let us define a relation by uRv iff u and v contain the same number of 1's.

Is R an equivalence relation? Why or why not?

2. Consider the set $S = \{1, 2, 3, 4\}$ with the subseteq relation $\subseteq$. We know that $(\mathcal{P}(S), \subseteq)$ is a poset.

- (a) Draw its Hasse diagram.
- (b) What is the length of the longest chain (chain with maximum possible number of elements) in this poset?
- (c) What is the size of the largest anti-chain in this poset?
- (d) Give two (different) topological sorts of the poset.

3. Consider the poset $(\mathbb{Z}^+, |)$, i.e., positive integers with divisibility ordering.

- (a) Give an example of a chain of length 5 in this poset.
- (b) Give an example of an anti-chain of length 5 in this poset.
- (c) Does this poset have:
 - i. a minimal element (an element in the poset such that there is no element smaller than it in the poset)
 - ii. a maximal element (an element in the poset such that there is no element larger than it in the poset)

- iii. a minimum or least element (i.e., it is smaller than all elements in the poset).
- iv. a maximum or greatest element (i.e., it is larger than all elements in the poset).
- v. an infinite chain (a chain with infinitely many elements)
- vi. an infinite anti-chain (an anti-chain with infinitely many elements)

For each of the above, if you claim there exists one, give an example, otherwise explain why there can't be any.

Part 2

4. A maximal chain is a chain that is not a subset of a larger chain. Prove or disprove: every maximal chain in a finite poset $(S, \preceq)$ contains a minimal element of S .
5. Consider a necklace made of 3 beads, each of which can be either red, white or blue. Let S be the set of all such necklaces. Define the following relation R on S as: $N_1 R N_2$ iff necklace N_2 can be obtained from necklace N_1 by rotating it (and *not* allowing to flip the necklace).
 - (a) Show that R is an equivalence relation.
 - (b) What are the equivalence classes of R ?
 - (c) Is the number of elements in each equivalence class the same? Is there a relationship between the number of elements in an equivalence class of R and the total number of elements in S ?
 - (d) If in the definition of the relation, we allow flipping of the necklace as well: that is, $N_1 R' N_2$ iff necklace N_2 can be obtained from necklace N_1 by rotating or flipping it. Is R' an equivalence relation? Why or why not?